

Tide retrospective: *Rescaled 2D moment asymptotic (V2)*

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Abstract

T1 (mixed even-power 2D moments) landed an *exact* closed form for $\langle x^{2j}y^{2k} \rangle_{2D,t}$ under the quartic–sextic potential. The candidate V1 paired with T1 left a follow-up labelled “generic- k J-function-style 2D moment asymptotic”; this Tide discharges it by packaging T1’s exact identity into three downstream- friendly forms: an algebraic reformulation isolating the t -power, a rescaled **Tendsto**, and an **Asymptotics.IsEquivalent**. No new analysis was required; T1 already supplied an exact identity, so the “asymptotic” packaging reduced to “a function eventually equal to a constant tends to that constant”, closed by **tendsto.congr’** + **tendsto.const_nhds**. 143 lines, single session, one spurious-**field_simp** fix needed for the exact lemma. Zero **sorry**, zero **axiom**, zero **native_decide**.

1 Setting

Two related framings circulated around V2 before Step 2. The survey direction said “joint scaling $t^{-(2j+1)/4-(2k+1)/6}$ as $t \rightarrow \infty$ ”, which describes the unnormalised numerator integral. The underlying moment, however, scales as $t^{-(j/2+k/3)}$ — T1 already gives this scaling *exactly*, not asymptotically, since T1’s RHS factors as $(24/t)^{j/2} \cdot \Gamma_1 \cdot (720/t)^{k/3} \cdot \Gamma_2$. So V2 is not a fresh asymptotic theorem; it’s the algebraic and asymptotic packaging of T1.

This matters: the survey wording inherited some confusion from T1’s follow-up section. Naming the unnormalised numerator and the moment got telescoped together, and the suggestion “combines **kth_jfunction_asymptotic** with T1’s factor theorem” is also misleading — N1’s J-function asymptotic requires a globally bounded prior, and a polynomial prior $\pi(x) = x^{2j}$ is unbounded, so N1 doesn’t apply directly to the numerator integrand. The correct route is via T1 (or via 1D **quartic_moment_even** + **sextic_moment_even** for the numerator separately), in either case producing an exact identity rather than an asymptotic.

2 The thing formalised

Three theorems in **Laplace.TwoD**:

1. **Exact reformulation.** For $t > 0$ and $j, k \in \mathbb{N}$,

$$\langle x^{2j}y^{2k} \rangle_{2D,t} = C(j, k) \cdot t^{-(j/2+k/3)}$$

where

$$C(j, k) = 24^{j/2} 720^{k/3} \frac{\Gamma((2j+1)/4)}{\Gamma(1/4)} \frac{\Gamma((2k+1)/6)}{\Gamma(1/6)}.$$

Named `gibbsExpectation_quarticSextic_pow_pow_eq_const_mul_rpow`. The constant prefactor is recorded as a named definition `quarticSexticMomentConst j k`.

2. **Rescaled Tendsto.** As $t \rightarrow \infty$,

$$t^{j/2+k/3} \langle x^{2j} y^{2k} \rangle_{2D,t} \rightarrow C(j, k).$$

Named `gibbsExpectation_quarticSextic_pow_pow_rescaled_tendsto`.

3. **IsEquivalent.** At `atTop`,

$$\langle x^{2j} y^{2k} \rangle_{2D,t} \sim C(j, k) \cdot t^{-(j/2+k/3)}.$$

Named `gibbsExpectation_quarticSextic_pow_pow_isEquivalent_rpow`.

3 Proof strategy

The exact reformulation is pure `Real.rpow` algebra. T1's RHS contains $(24/t)^{j/2}$ and $(720/t)^{k/3}$; using `Real.div_rpow` (for $a/b)^x = a^x/b^x$ with $a, b \geq 0$) splits each into a base power and an inverse- t power. Combining the two inverse- t pieces uses `Real.rpow_add` (valid for $t > 0$) to write $t^{-(j/2+k/3)} = t^{-j/2} \cdot t^{-k/3}$, then `Real.rpow_neg` converts negative-exponent `rpow` to the reciprocal form. After those rewrites, `ring` closes the goal — both sides are the same polynomial in the named $24^{j/2}, 720^{k/3}, \Gamma$ -ratios, $t^{j/2}, t^{k/3}$.

The rescaled `Tendsto` is the clean GPT idiom: the rescaled function is *eventually equal* (on $\{t > 0\}$) to a constant function. Concretely, $t^{j/2+k/3}$ times the moment equals $C(j, k) \cdot t^{j/2+k/3} \cdot t^{-(j/2+k/3)}$, and the t -factors annihilate via `Real.rpow_add`, leaving $C(j, k)$. The `Tendsto` then follows from `tendsto_const_nhds` via `tendsto_congr'`.

The `IsEquivalent` is the cleanest of all: `refl` modulo eventual-equality. The two functions differ only by “ $\langle \dots \rangle$ versus $C(j, k) \cdot t^{-(j/2+k/3)}$ ”; the exact reformulation says these are pointwise equal on $\{t > 0\}$ and so eventually equal at `atTop`.

4 Roadblocks and resolutions

One roadblock: my first draft of the exact reformulation closed with a `field.simp` followed by `ring`, and the build errored with “No goals to be solved” at the `ring`. `field.simp` had already closed the goal after the `rpow_add + rpow_neg` normalisations. Removing the `field.simp` left `ring` to do the work directly, and the build went green. Pattern worth recording: after explicit `rpow_add/rpow_neg` rewrites that already normalise the t -structure of the goal, `ring` alone tends to close; `field.simp` pre-processing is redundant and can produce a goalless state that breaks the subsequent tactic.

No GPT mid-proof consult was needed. The pre-proof consult was the load-bearing one; once the choice of three-theorem packaging and the eventual-equality idiom were set, the implementation was mechanical.

5 What was Mathlib, what was new

From Mathlib. `Real.div_rpow` (for splitting $(a/b)^x$), `Real.rpow_add` (for combining $t^a \cdot t^b = t^{a+b}$ when $t > 0$), `Real.rpow_neg` (for converting t^{-x} to $(t^x)^{-1}$ when $t \geq 0$), `tendsto_const_nhds` and `Tendsto.congr'`, `Asymptotics.IsEquivalent.refl` and `IsEquivalent.congr_left`, plus standard `ring`.

From the seabed. Only T1’s exact closed form `gibbsExpectation.quarticSextic.pow_pow.eq`. Nothing else from the laplace seabed was load-bearing.

New here. Three theorems + the named constant `quarticSexticMomentConst`. The architectural contribution is the realisation, surfaced during Step 2 deliberation, that V2’s “asymptotic” framing was conceptually redundant against T1’s exact identity — and the corresponding decision to ship the *exact reformulation* as a keystone with the two asymptotic packagings as five-line corollaries of it.

6 Lessons

Exact identities expose asymptotics for free. Whenever a seabed lemma already gives a closed-form equality (rather than an asymptotic estimate), every downstream “asymptotic” theorem about it is just a packaging of the same exact identity in a different filter-theoretic shape. Don’t reach for `isEquivalent_iff_tendsto_one`-style ratio arguments; the cleanest proof is via `tendsto_congr`’ on the eventually-constant rescaled function. GPT-5.5 Pro flagged this idiom at Step 2; worth recording as the standard shape for “ $\langle \dots \rangle = C \cdot t^{-\alpha}$ ”-style packagings.

Inherit the keystone-plus-corollary structure when in doubt. L2 (J-function asymptotic equivalence) is the closest exemplar: it shipped the ratio convergence and the `IsEquivalent` together. V2 inherited this dual-form pattern, adding the algebraic-reformulation lemma as a third keystone. The result is three theorems for the price of one proof’s work: once the exact reformulation is in place, the other two are `tendsto_congr`’ away.

field_simp can close goals already normalised by rpow rewrites. A spurious `field_simp` after `Real.rpow_add/rpow_neg` normalisations can succeed silently, leaving a subsequent `ring` with no goals. Drop the `field_simp` if `ring` fails with “No goals to be solved”.

Question survey-wording before deliberation. The V2 survey direction had two layers of conceptual drift inherited from T1’s follow-up: “J-function-style asymptotic” was actually unreachable from N1’s machinery (polynomial prior unbounded), and “joint scaling $t^{-(2j+1)/4-(2k+1)/6}$ ” described the unnormalised numerator, not the moment. Surfacing both layers in the tide-log’s seabed-snapshot section let Step 2 deliberate cleanly on the right object.

7 Follow-ups

- *Numerator-side asymptotic* (the survey’s actual literal statement). Build the analogous packaging for $\iint x^{2j} y^{2k} e^{-tL(x,y)} dx dy$ scaling as $t^{-(2j+1)/4-(2k+1)/6}$. Requires factoring the 2D numerator via separability (analogous to A2’s partition-factor theorem but for the numerator) and applying the 1D `quartic_moment_even` / `sextic_moment_even` closed forms. ~40 lines. Standalone follow-up tide.
- *Odd-side “asymptotic”*. V1 proves the odd-moment 2D vanishing as an exact identity. The asymptotic packaging would be the trivial $\langle \dots \rangle = 0 \sim 0$, presumably useless. Skip.
- *Promote `quarticSexticMomentConst` to lean-common*. Only if a third consumer materialises; currently the named constant has one use site (this file). Below the refactor threshold.

- `\leanref-tag the primer/grammar` paper. Pairs with K2/U3 paper-tagging backlog once the primer’s 2D-moment-asymptotic statements have a TeX home.

Acknowledgements

GPT-5.5 Pro’s deliberation contributed: the vote for Candidate C (both the `Tendsto` and the `IsEquivalent`, with the exact reformulation as a stepping stone) over the user’s simpler Candidate A; the `tendsto_congr`’ / eventual-equality idiom as the right proof shape for the asymptotic packagings; the explicit rejection of Candidate D (the unnormalised numerator) as a different object; and the naming triplet (`_eq_const_mul_rpow`, `_rescaled_tendsto`, `_isEquivalent_rpow`). Full consult preserved in the primer’s tide-log GPT-responses folder.